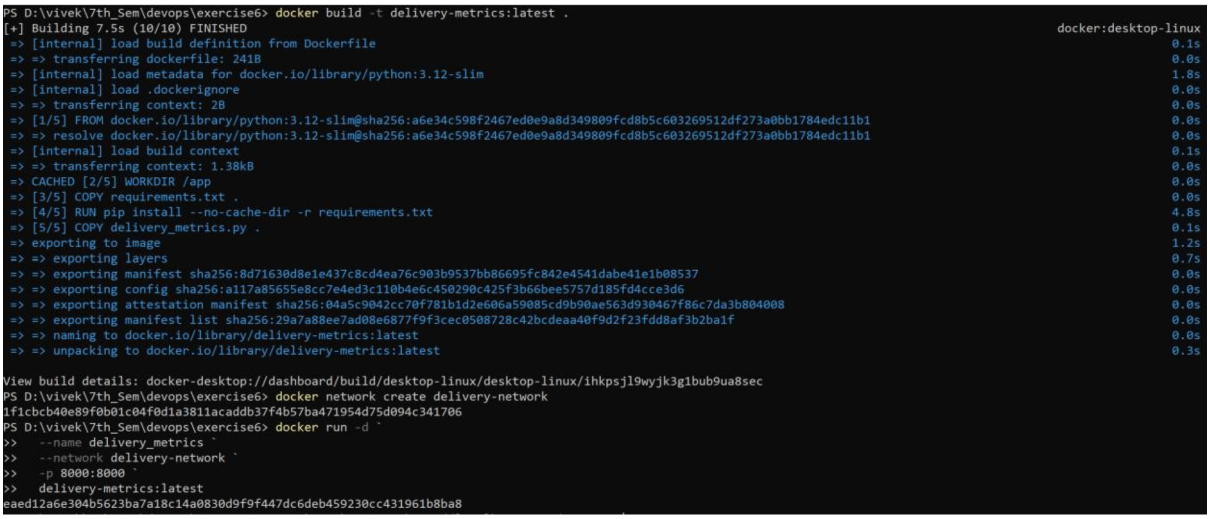
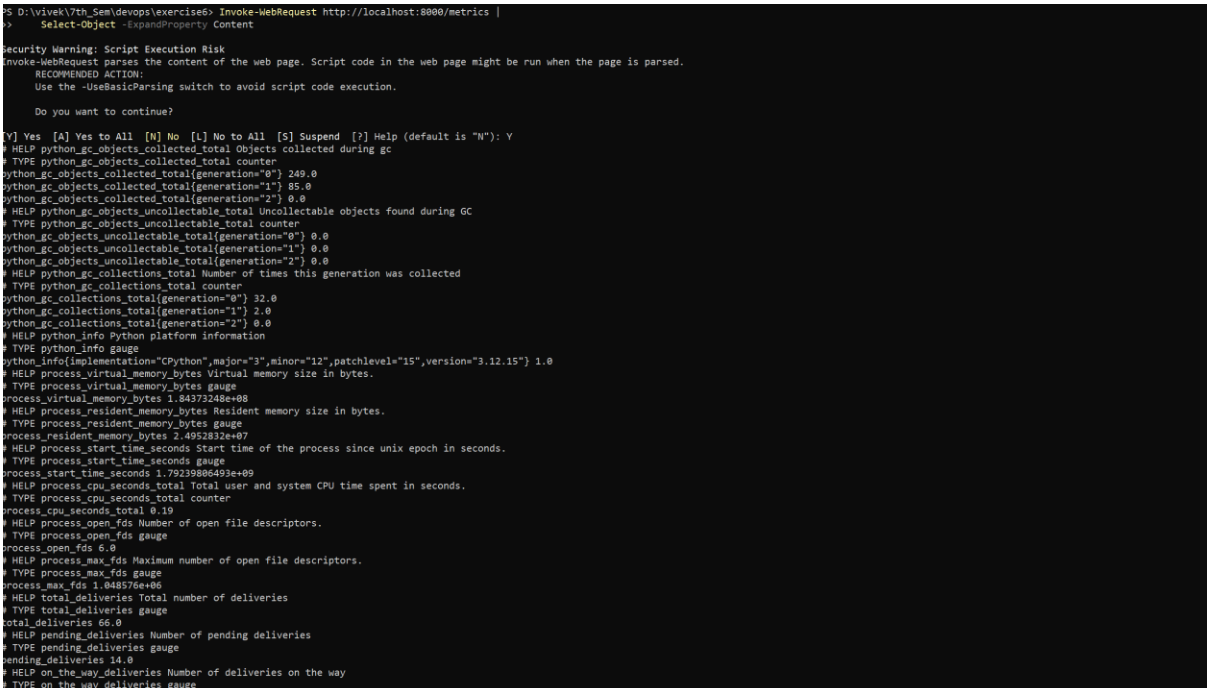
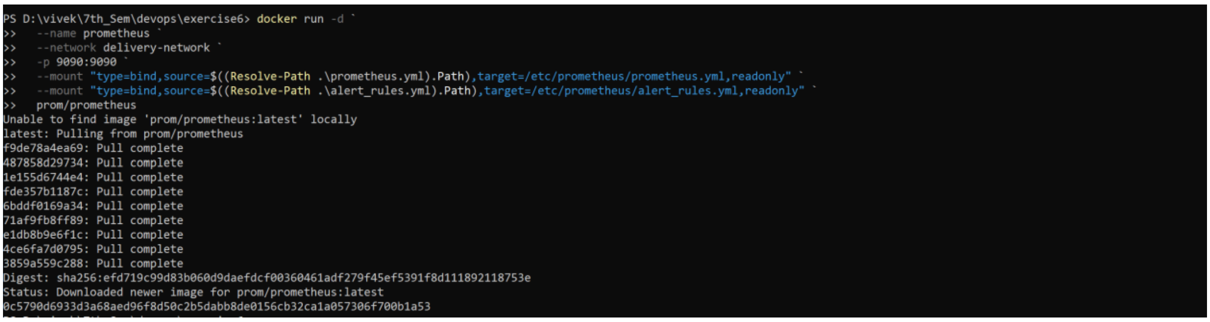
Devops – Exercise 6



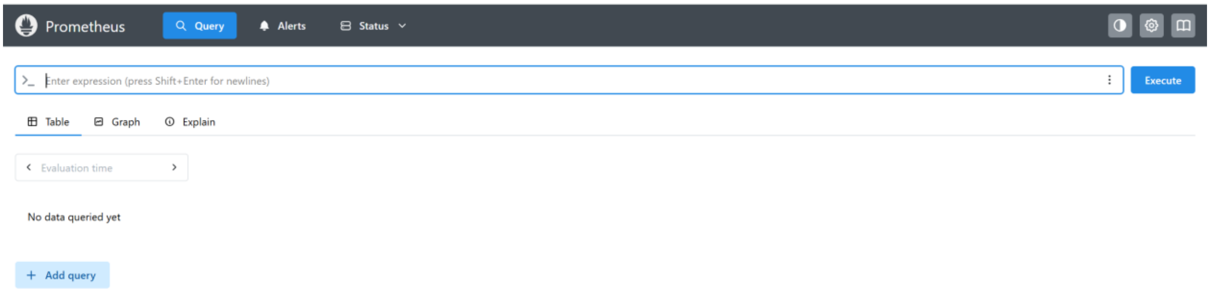
```
PS D:\vivek\7th_Sem\devops\exercise6> docker build -t delivery-metrics:latest .
[+] Building 7.5s (10/10) FINISHED                                                                                                                docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                                                              0.1s
 => => transferring dockerfile: 241B                                                                                                                              0.0s
 => [internal] load metadata for docker.io/library/python:3.12-slim                                                                                               1.8s
 => [internal] load .dockerignore                                                                                                                                 0.0s
 => => transferring context: 2B                                                                                                                                   0.0s
 => [1/5] FROM docker.io/library/python:3.12-slim@sha256:a6e34c598f2467ed0e9a8d349809fcd8b5c603269512df273a0bb1784edc11b1                                         0.0s
 => => resolve docker.io/library/python:3.12-slim@sha256:a6e34c598f2467ed0e9a8d349809fcd8b5c603269512df273a0bb1784edc11b1                                         0.0s
 => [internal] load build context                                                                                                                                 0.1s
 => => transferring context: 1.38kB                                                                                                                               0.0s
 => CACHED [2/5] WORKDIR /app                                                                                                                                     0.0s
 => [3/5] COPY requirements.txt .                                                                                                                                 0.0s
 => [4/5] RUN pip install --no-cache-dir -r requirements.txt                                                                                                      4.8s
 => [5/5] COPY delivery_metrics.py .                                                                                                                              0.1s
 => exporting to image                                                                                                                                            1.2s
 => => exporting layers                                                                                                                                           0.7s
 => => exporting manifest sha256:8d71630d8e1e437c8cd4ea76c903b9537bb86695fc842e4541dabe41e1b08537                                                                 0.0s
 => => exporting config sha256:a117a85655e8cc7e4ed3c110b4e6c450290c425f3b66bee5757d185fd4cce3d6                                                                   0.0s
 => => exporting attestation manifest sha256:04a5c9042cc70f781b1d2e606a59085cd9b90ae563d930467f86c7da3b804008                                                     0.0s
 => => exporting manifest list sha256:29a7a88ee7ad08e6877f9f3cec0508728c42bcdeaa40f9d2f23fdd8af3b2ba1f                                                            0.0s
 => => naming to docker.io/library/delivery-metrics:latest                                                                                                        0.0s
 => => unpacking to docker.io/library/delivery-metrics:latest                                                                                                     0.3s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/ihkpsjl9wyjk3g1bub9ua8sec
PS D:\vivek\7th_Sem\devops\exercise6> docker network create delivery-network
1f1cbcb40e89f0b01c04f0d1a3811acaddb37f4b57ba471954d75d094c341706
PS D:\vivek\7th_Sem\devops\exercise6> docker run -d `
>>   --name delivery_metrics `
>>   --network delivery-network `
>>   -p 8000:8000 `
>>   delivery-metrics:latest
eaed12a6e304b5623ba7a18c14a0830d9f9f447dc6deb459230cc431961b8ba8
```



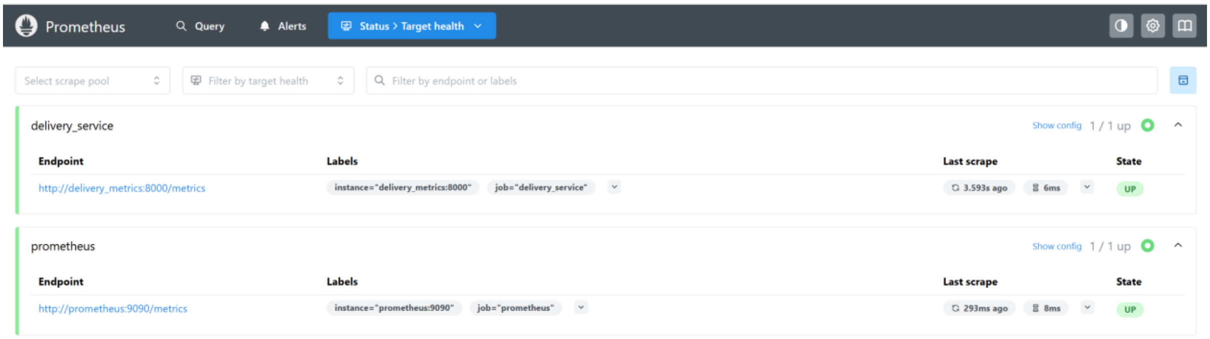
```
PS D:\vivek\7th_Sem\devops\exercise6> Invoke-WebRequest http://localhost:8000/metrics |
>>     Select-Object -ExpandProperty Content

Security Warning: Script Execution Risk
Invoke-WebRequest parses the content of the web page. Script code in the web page might be run when the page is parsed.
      RECOMMENDED ACTION:
      Use the -UseBasicParsing switch to avoid script code execution.

      Do you want to continue?

[Y] Yes  [A] Yes to All  [N] No  [L] No to All  [S] Suspend  [?] Help (default is "N"): Y
# HELP python_gc_objects_collected_total Objects collected during gc
# TYPE python_gc_objects_collected_total counter
python_gc_objects_collected_total{generation="0"} 249.0
python_gc_objects_collected_total{generation="1"} 85.0
python_gc_objects_collected_total{generation="2"} 0.0
# HELP python_gc_objects_uncollectable_total Uncollectable objects found during GC
# TYPE python_gc_objects_uncollectable_total counter
python_gc_objects_uncollectable_total{generation="0"} 0.0
python_gc_objects_uncollectable_total{generation="1"} 0.0
python_gc_objects_uncollectable_total{generation="2"} 0.0
# HELP python_gc_collections_total Number of times this generation was collected
# TYPE python_gc_collections_total counter
python_gc_collections_total{generation="0"} 32.0
python_gc_collections_total{generation="1"} 2.0
python_gc_collections_total{generation="2"} 0.0
# HELP python_info Python platform information
# TYPE python_info gauge
python_info{implementation="CPython",major="3",minor="12",patchlevel="15",version="3.12.15"} 1.0
# HELP process_virtual_memory_bytes Virtual memory size in bytes.
# TYPE process_virtual_memory_bytes gauge
process_virtual_memory_bytes 1.84373248e+08
# HELP process_resident_memory_bytes Resident memory size in bytes.
# TYPE process_resident_memory_bytes gauge
process_resident_memory_bytes 2.4952832e+07
# HELP process_start_time_seconds Start time of the process since unix epoch in seconds.
# TYPE process_start_time_seconds gauge
process_start_time_seconds 1.79239806493e+09
# HELP process_cpu_seconds_total Total user and system CPU time spent in seconds.
# TYPE process_cpu_seconds_total counter
process_cpu_seconds_total 0.19
# HELP process_open_fds Number of open file descriptors.
# TYPE process_open_fds gauge
process_open_fds 6.0
# HELP process_max_fds Maximum number of open file descriptors.
# TYPE process_max_fds gauge
process_max_fds 1.048576e+06
# HELP total_deliveries Total number of deliveries
# TYPE total_deliveries gauge
total_deliveries 66.0
# HELP pending_deliveries Number of pending deliveries
# TYPE pending_deliveries gauge
pending_deliveries 14.0
# HELP on_the_way_deliveries Number of deliveries on the way
# TYPE on_the_way_deliveries gauge
```



```
PS D:\vivek\7th_Sem\devops\exercise6> docker run -d `
>>   --name prometheus `
>>   --network delivery-network `
>>   -p 9090:9090 `
>>   --mount "type=bind,source=$((Resolve-Path .\prometheus.yml).Path),target=/etc/prometheus/prometheus.yml,readonly" `
>>   --mount "type=bind,source=$((Resolve-Path .\alert_rules.yml).Path),target=/etc/prometheus/alert_rules.yml,readonly" `
>>   prom/prometheus
Unable to find image 'prom/prometheus:latest' locally
latest: Pulling from prom/prometheus
f9de78a4ea69: Pull complete
487858d29734: Pull complete
1e155d6744e4: Pull complete
fde357b1187c: Pull complete
6bddf0169a34: Pull complete
71af9fb8ff89: Pull complete
e1db8b9e6f1c: Pull complete
4ce6fa7d0795: Pull complete
3859a559c288: Pull complete
Digest: sha256:efd719c99d83b060d9daefdcf00360461adf279f45ef5391f8d111892118753e
Status: Downloaded newer image for prom/prometheus:latest
0c5790d6933d3a68aed96f8d50c2b5dabb8de0156cb32ca1a057306f700b1a53
```



No data queried yet

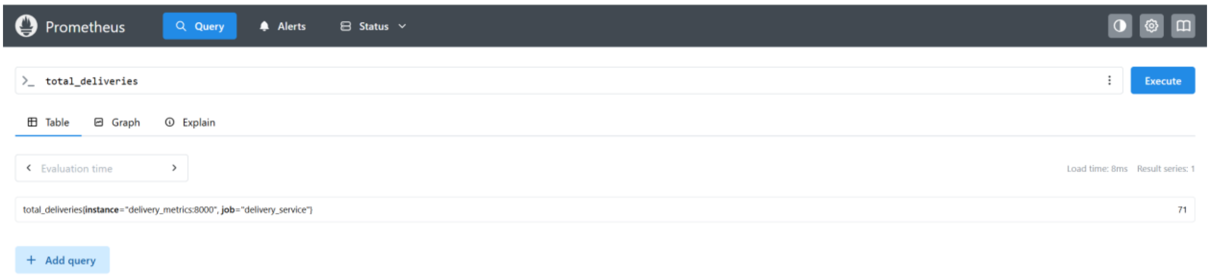


delivery_service 1 / 1 up

Endpoint	Labels	Last scrape	State
http://delivery_metrics:8000/metrics	instance="delivery_metrics:8000" job="delivery_service"	3.593s ago 6ms	UP

prometheus 1 / 1 up

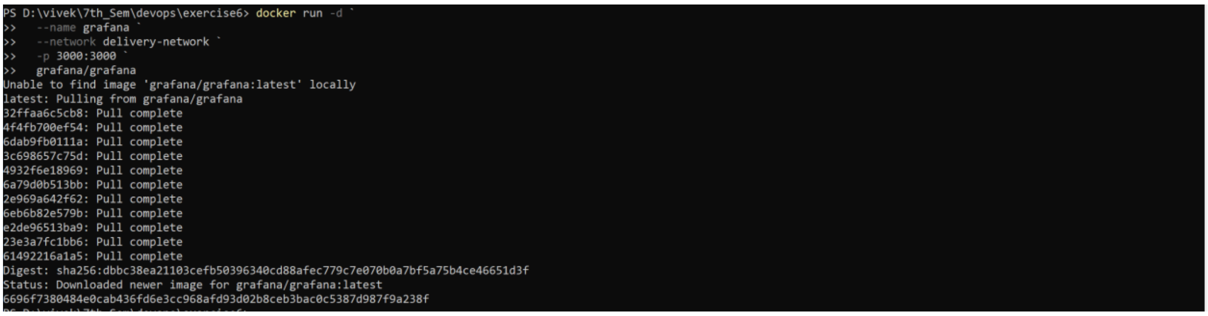
Endpoint	Labels	Last scrape	State
http://prometheus:9090/metrics	instance="prometheus:9090" job="prometheus"	293ms ago 8ms	UP



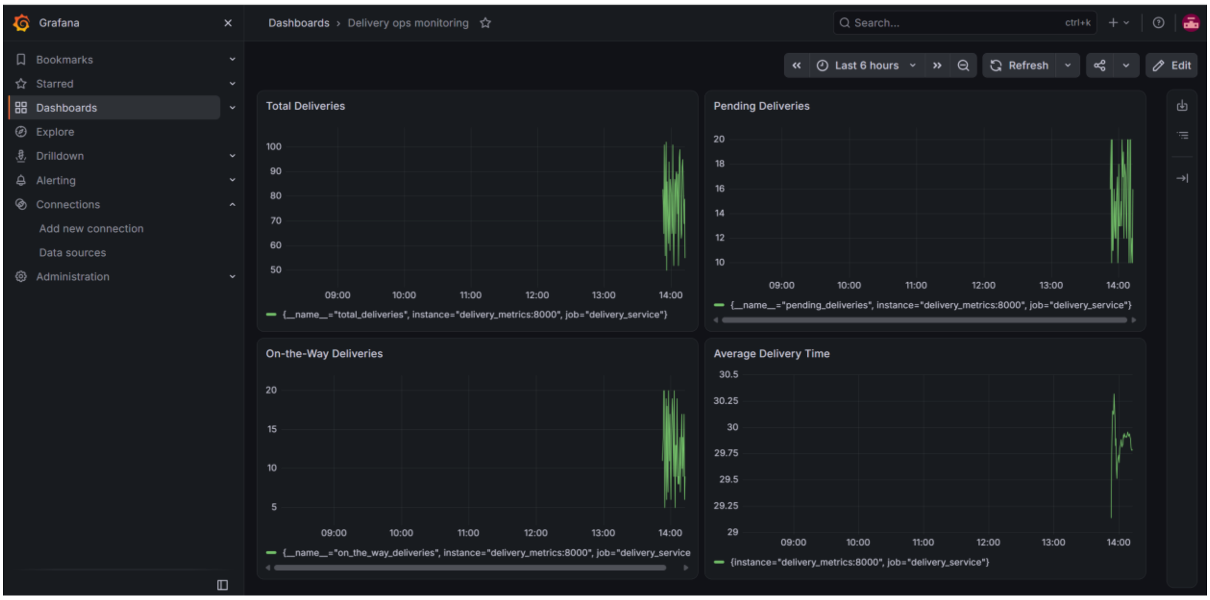
```
total_deliveries
```

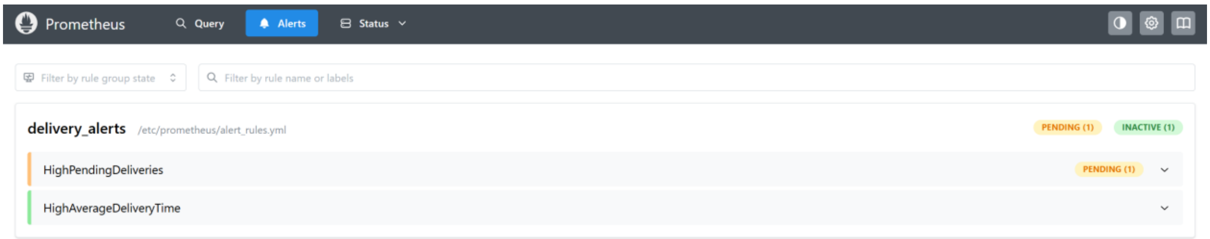
Load time: 8ms Result series: 1

total_deliveries{**instance**="delivery_metrics:8000", **job**="delivery_service"}	71



```
PS D:\vivek\7th_Sem\devops\exercise6> docker run -d `
>>   --name grafana `
>>   --network delivery-network `
>>   -p 3000:3000 `
>>   grafana/grafana
Unable to find image 'grafana/grafana:latest' locally
latest: Pulling from grafana/grafana
32ffaa6c5cb8: Pull complete
4f4fb700ef54: Pull complete
6dab9fb0111a: Pull complete
3c698657c75d: Pull complete
4932f6e18969: Pull complete
6a79d0b513bb: Pull complete
2e969a642f62: Pull complete
6eb6b82e579b: Pull complete
e2de96513ba9: Pull complete
23e3a7fc1bb6: Pull complete
61492216a1a5: Pull complete
Digest: sha256:dbbc38ea21103cefb50396340cd88afec779c7e070b0a7bf5a75b4ce46651d3f
Status: Downloaded newer image for grafana/grafana:latest
6696f7380484e0cab436fd6e3cc968afd93d02b8ceb3bac0c5387d987f9a238f
```

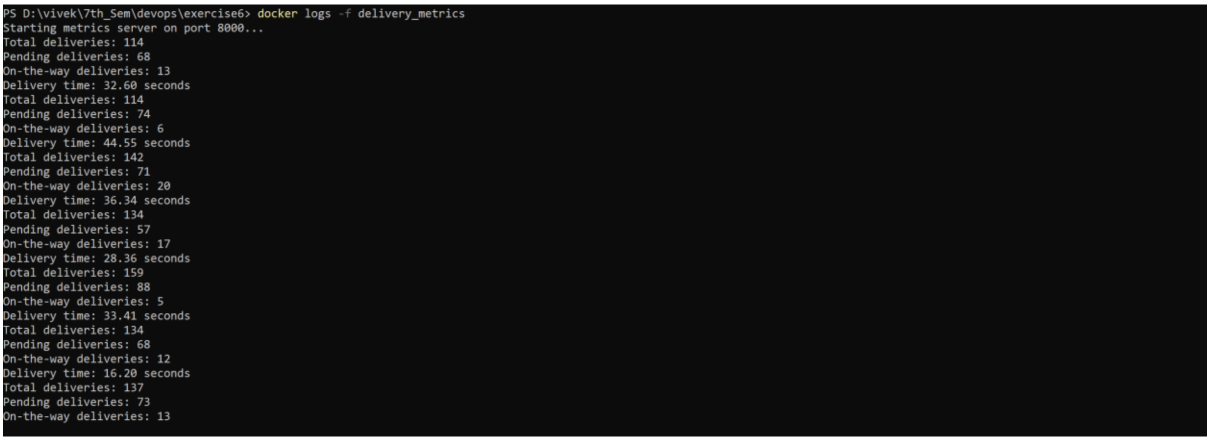




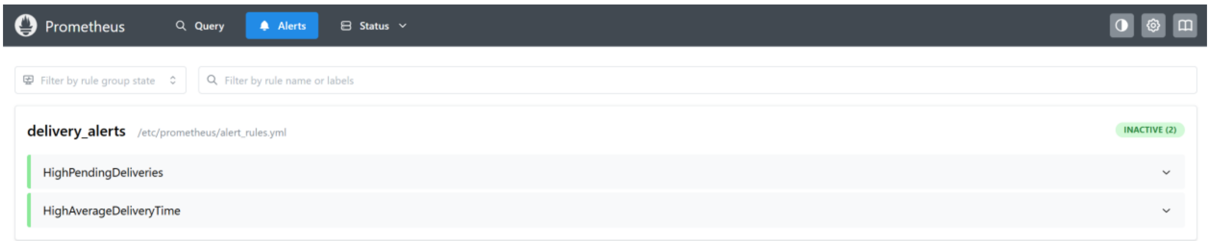
delivery_alerts /etc/prometheus/alert_rules.yml PENDING (1) INACTIVE (1)

HighPendingDeliveries PENDING (1)

HighAverageDeliveryTime



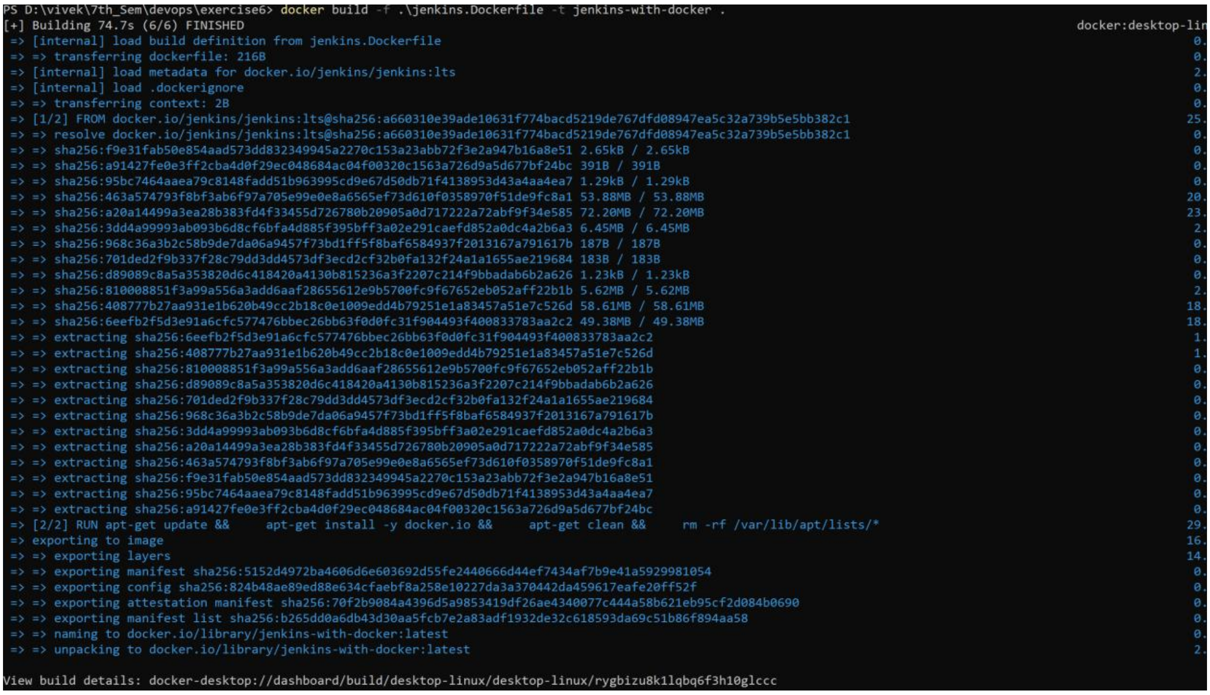
```
PS D:\vivek\7th_Sem\devops\exercise6> docker logs -f delivery_metrics
Starting metrics server on port 8000...
Total deliveries: 114
Pending deliveries: 68
On-the-way deliveries: 13
Delivery time: 32.60 seconds
Total deliveries: 114
Pending deliveries: 74
On-the-way deliveries: 6
Delivery time: 44.55 seconds
Total deliveries: 142
Pending deliveries: 71
On-the-way deliveries: 20
Delivery time: 36.34 seconds
Total deliveries: 134
Pending deliveries: 57
On-the-way deliveries: 17
Delivery time: 28.36 seconds
Total deliveries: 159
Pending deliveries: 88
On-the-way deliveries: 5
Delivery time: 33.41 seconds
Total deliveries: 134
Pending deliveries: 68
On-the-way deliveries: 12
Delivery time: 16.20 seconds
Total deliveries: 137
Pending deliveries: 73
On-the-way deliveries: 13
```



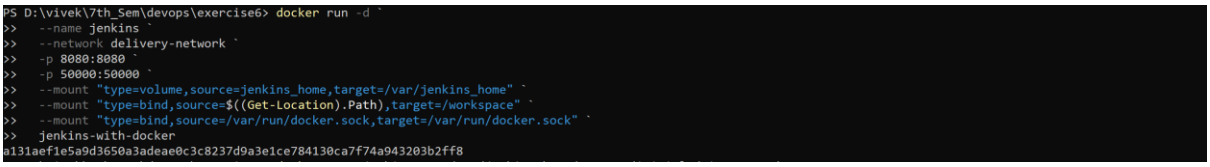
delivery_alerts /etc/prometheus/alert_rules.yml INACTIVE (2)

HighPendingDeliveries

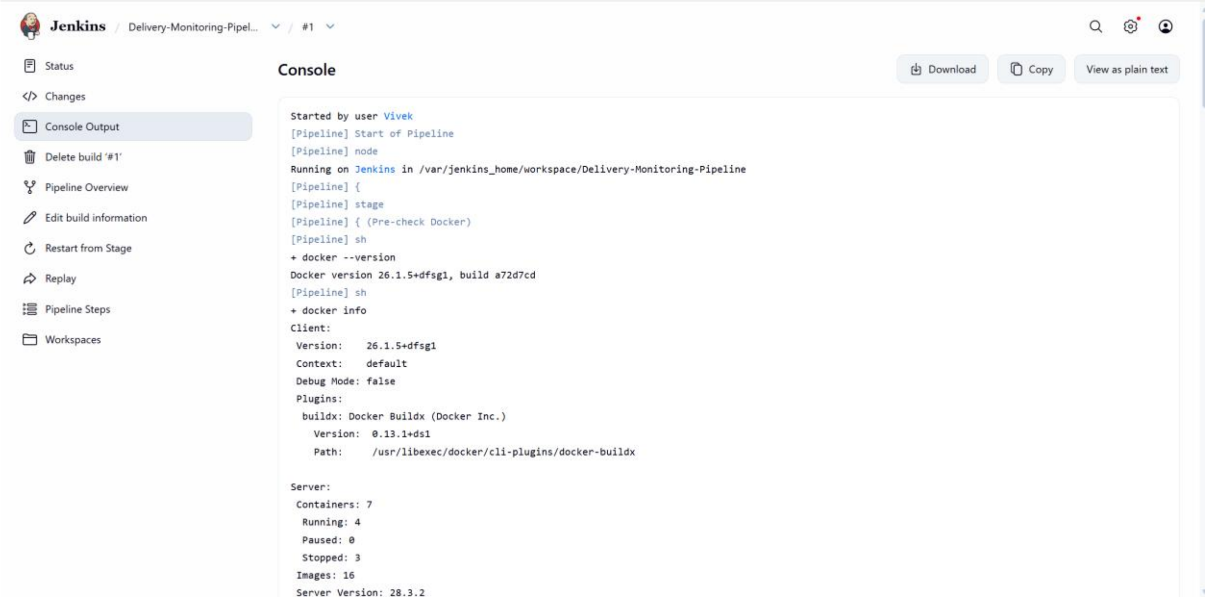
HighAverageDeliveryTime



```
PS D:\vivek\7th_Sem\devops\exercise6> docker build -f .\jenkins.Dockerfile -t jenkins-with-docker .
[+] Building 74.7s (6/6) FINISHED                                                                                                                      docker:desktop-lin
 => [internal] load build definition from jenkins.Dockerfile                                                                                                          0.
 => => transferring dockerfile: 216B                                                                                                                                  0.
 => [internal] load metadata for docker.io/jenkins/jenkins:lts                                                                                                        2.
 => [internal] load .dockerignore                                                                                                                                     0.
 => => transferring context: 2B                                                                                                                                       0.
 => [1/2] FROM docker.io/jenkins/jenkins:lts@sha256:a660310e39ade10631f774bacd5219de767dfd08947ea5c32a739b5e5bb382c1                                                 25.
 => => resolve docker.io/jenkins/jenkins:lts@sha256:a660310e39ade10631f774bacd5219de767dfd08947ea5c32a739b5e5bb382c1                                                  0.
 => => sha256:f9e31fab50e854aad573dd832349945a2270c153a23abb72f3e2a947b16a8e51 2.65kB / 2.65kB                                                                        0.
 => => sha256:a91427fe0e3ff2cba4d0f29ec048684ac04f00320c1563a726d9a5d677bf24bc 391B / 391B                                                                            0.
 => => sha256:95bc7464aaea79c8148fadd51b963995cd9e67d50db71f4138953d43a4aa4ea7 1.29kB / 1.29kB                                                                        0.
 => => sha256:463a574793f8bf3ab6f97a705e99e0e8a6565ef73d610f0358970f51de9fc8a1 53.88MB / 53.88MB                                                                     20.
 => => sha256:a20a14499a3ea28b383fd4f33455d726780b20905a0d717222a72abf9f34e585 72.20MB / 72.20MB                                                                     23.
 => => sha256:3dd4a99993ab093b6d8cf6bfa4d885f395bff3a02e291caefd852a0dc4a2b6a3 6.45MB / 6.45MB                                                                        2.
 => => sha256:968c36a3b2c58b9de7da06a9457f73bd1ff5f8baf6584937f2013167a791617b 187B / 187B                                                                            0.
 => => sha256:701ded2f9b337f28c79dd3dd4573df3ecd2cf32b0fa132f24a1a1655ae219684 183B / 183B                                                                            0.
 => => sha256:d89089c8a5a353820d6c418420a4130b815236a3f2207c214f9bbadab6b2a626 1.23kB / 1.23kB                                                                        0.
 => => sha256:810008851f3a99a556a3add6aaf28655612e9b5700fc9f67652eb052aff22b1b 5.62MB / 5.62MB                                                                        2.
 => => sha256:408777b27aa931e1b620b49cc2b18c0e1009edd4b79251e1a83457a51e7c526d 58.61MB / 58.61MB                                                                     18.
 => => sha256:6eefb2f5d3e91a6cfc577476bbec26bb63f0d0fc31f904493f400833783aa2c2 49.38MB / 49.38MB                                                                     18.
 => => extracting sha256:6eefb2f5d3e91a6cfc577476bbec26bb63f0d0fc31f904493f400833783aa2c2                                                                             1.
 => => extracting sha256:408777b27aa931e1b620b49cc2b18c0e1009edd4b79251e1a83457a51e7c526d                                                                             1.
 => => extracting sha256:810008851f3a99a556a3add6aaf28655612e9b5700fc9f67652eb052aff22b1b                                                                             0.
 => => extracting sha256:d89089c8a5a353820d6c418420a4130b815236a3f2207c214f9bbadab6b2a626                                                                             0.
 => => extracting sha256:701ded2f9b337f28c79dd3dd4573df3ecd2cf32b0fa132f24a1a1655ae219684                                                                             0.
 => => extracting sha256:968c36a3b2c58b9de7da06a9457f73bd1ff5f8baf6584937f2013167a791617b                                                                             0.
 => => extracting sha256:3dd4a99993ab093b6d8cf6bfa4d885f395bff3a02e291caefd852a0dc4a2b6a3                                                                             0.
 => => extracting sha256:a20a14499a3ea28b383fd4f33455d726780b20905a0d717222a72abf9f34e585                                                                             0.
 => => extracting sha256:463a574793f8bf3ab6f97a705e99e0e8a6565ef73d610f0358970f51de9fc8a1                                                                             0.
 => => extracting sha256:f9e31fab50e854aad573dd832349945a2270c153a23abb72f3e2a947b16a8e51                                                                             0.
 => => extracting sha256:95bc7464aaea79c8148fadd51b963995cd9e67d50db71f4138953d43a4aa4ea7                                                                             0.
 => => extracting sha256:a91427fe0e3ff2cba4d0f29ec048684ac04f00320c1563a726d9a5d677bf24bc                                                                             0.
 => [2/2] RUN apt-get update &&     apt-get install -y docker.io &&     apt-get clean &&     rm -rf /var/lib/apt/lists/*                                             29.
 => exporting to image                                                                                                                                               16.
 => => exporting layers                                                                                                                                              14.
 => => exporting manifest sha256:5152d4972ba4606d6e603692d55fe2440666d44ef7434af7b9e41a5929981054                                                                     0.
 => => exporting config sha256:824b48ae89ed88e634cfaebf8a258e10227da3a370442da459617eafe20ff52f                                                                       0.
 => => exporting attestation manifest sha256:70f2b9084a4396d5a9853419df26ae4340077c444a58b621eb95cf2d084b0690                                                         0.
 => => exporting manifest list sha256:b265dd0a6db43d30aa5fcb7e2a83adf1932de32c618593da69c51b86f894aa58                                                                0.
 => => naming to docker.io/library/jenkins-with-docker:latest                                                                                                         0.
 => => unpacking to docker.io/library/jenkins-with-docker:latest                                                                                                      2.

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/rygbizu8k1lqbq6f3h10glccc
```



```
PS D:\vivek\7th_Sem\devops\exercise6> docker run -d `
>>   --name jenkins `
>>   --network delivery-network `
>>   -p 8080:8080 `
>>   -p 50000:50000 `
>>   --mount "type=volume,source=jenkins_home,target=/var/jenkins_home" `
>>   --mount "type=bind,source=$((Get-Location).Path),target=/workspace" `
>>   --mount "type=bind,source=/var/run/docker.sock,target=/var/run/docker.sock" `
>>   jenkins-with-docker
a131aef1e5a9d3650a3adeae0c3c8237d9a3e1ce784130ca7f74a943203b2ff8
```



```
Started by user Vivek
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/jenkins_home/workspace/Delivery-Monitoring-Pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Pre-check Docker)
[Pipeline] sh
+ docker --version
Docker version 26.1.5+dfsg1, build a72d7cd
[Pipeline] sh
+ docker info
Client:
 Version:    26.1.5+dfsg1
 Context:    default
 Debug Mode: false
 Plugins:
  buildx: Docker Buildx (Docker Inc.)
    Version:  0.13.1+ds1
    Path:     /usr/libexec/docker/cli-plugins/docker-buildx

Server:
 Containers: 7
  Running: 4
  Paused: 0
  Stopped: 3
 Images: 16
 Server Version: 28.3.2
```